

I225E Statistical Signal Processing

12. Linear Model

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Linear model

- Prediction and smoothing
- Non-causal and causal Wiener filter
- (Innovation filter and Laplace transform)
- Linear minimum mean squared error (LMMSE) prediction
- Extension to vector form

1. Signal Processing

Problem: Using stochastic process $X(\xi)$, ($a \leq \xi \leq b$), estimate the value of stochastic process $S(t)$ as

$$\hat{S}(t) = \int_a^b h(\alpha)X(\alpha)d\alpha$$

- **Smoothing**, in case estimation time t is within the data interval ($t \in [a, b]$).
- **Prediction**, in case estimation time t is outside of the data interval ($t \notin [a, b]$) and $X(t) = S(t)$.
- **Smoothing and Filtering**, in case estimation time t is outside of the data interval ($t \notin [a, b]$) and $X(t) \neq S(t)$.

Exercise 1

$S(t)$ is a stationary process.

Predict the future state $S(t + \tau)$ using the past state $S(t)$ as

$$\hat{S}(t + \tau) = aS(t).$$

Find the optimal coefficient a .

By orthogonality principle,

$$E\{(S(t + \tau) - aS(t))S(t)\} = 0$$

$$E\{S(t + \tau)S(t)\} - aE\{S(t)S(t)\} = 0$$

$$R_{SS}(\tau) - aR_{SS}(0) = 0$$

Hence,
$$a = \frac{R_{SS}(\tau)}{R_{SS}(0)}$$

Corresponding mean square error is,

$$\begin{aligned} \text{MSE} &= E\{(S(t + \tau) - aS(t))S(t + \tau)\} \\ &= E\{S(t + \tau)S(t + \tau)\} - aE\{S(t)S(t + \tau)\} \\ &= R_{SS}(0) - aR_{SS}(\tau) \\ &= \frac{R_{SS}^2(0) - R_{SS}^2(\tau)}{R_{SS}(0)} \end{aligned}$$

Exercise 2

Smoothing the present state of $S(t)$ using the present state of another process $X(t)$ as

$$\hat{S}(t) = aX(t).$$

Find the optimal coefficient a .

By orthogonality principle,

$$E\{(S(t) - aX(t))X(t)\} = 0$$

$$E\{S(t)X(t)\} - aE\{X(t)X(t)\} = 0$$

$$R_{SX}(0) - aR_{XX}(0) = 0$$

Hence,
$$a = \frac{R_{SX}(0)}{R_{XX}(0)}$$

Corresponding mean square error is,

$$\begin{aligned} \text{MSE} &= E\{(S(t) - aX(t))S(t)\} \\ &= E\{S(t)S(t)\} - aE\{X(t)S(t)\} \\ &= R_{SS}(0) - aR_{SX}(0) \\ &= \frac{R_{SS}(0)R_{XX}(0) - R_{SX}^2(0)}{R_{XX}(0)} \end{aligned}$$

2. Smoothing

■ Problem

Using the entire interval ($t \in [-\infty, \infty]$) of the following process

$$X(t) = S(t) + v(t),$$

estimate the present state of $S(t)$ as

$$\hat{S}(t) = \int_{-\infty}^{\infty} h(\alpha) X(t - \alpha) d\alpha.$$

Find the optimal filter $h(\alpha)$.

■ Solution

By orthogonality principle,

$$S(t) - \hat{S}(t) \perp X(\xi), \text{ for all } \xi \in [-\infty, \infty].$$

Denote $\xi = t - \tau$, For all τ

$$E\left\{\left[S(t) - \int_{-\infty}^{\infty} h(\alpha)X(t - \alpha)d\alpha\right]X(t - \tau)\right\} = 0$$

$$\begin{aligned} E\{S(t)X(t - \tau)\} - \int_{-\infty}^{\infty} h(\alpha)E\{X(t - \alpha)X(t - \tau)\}d\alpha \\ = 0 \end{aligned}$$

$$R_{SX}(\tau) - \int_{-\infty}^{\infty} h(\alpha)R_{XX}(\tau - \alpha)d\alpha = 0$$

(Wiener-Hopf equation)



Fourier transform of $R_{SX}(\tau) = h(\tau) * R_{XX}(\tau)$ gives

$$S_{SX}(\omega) = H(\omega)S_{XX}(\omega).$$

Hence,

$$H(\omega) = \frac{S_{SX}(\omega)}{S_{XX}(\omega)}.$$

This is called ***Non-causal Wiener Filter***.

Corresponding mean square error is

$$\begin{aligned} Bmse &= E\left\{\left(S(t) - \int_{-\infty}^{\infty} h(\alpha)X(t - \alpha)d\alpha\right)S(t)\right\} \\ &= E\{S(t)S(t)\} - \int_{-\infty}^{\infty} h(\alpha)E\{X(t - \alpha)S(t)\}d\alpha \\ &= R_{SS}(0) - \int_{-\infty}^{\infty} h(\alpha)R_{XS}(-\alpha)d\alpha \\ &= R_{SS}(\tau) - \int_{-\infty}^{\infty} h(\alpha)R_{XS}(\tau - \alpha)d\alpha|_{\tau=0} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} [S_{SS}(\omega) - H(\omega)S_{XS}(\omega)]e^{i\omega\tau}d\omega|_{\tau=0} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_{SS}(\omega)S_{XX}(\omega) - S_{SX}^2(\omega)}{S_{XX}(\omega)}d\omega \end{aligned}$$

Case Study: Signal Orthogonal to Noise

In case that process $S(t)$ is orthogonal to noise $v(t)$.

$$R_{Sv}(\tau) = 0.$$

This leads to

$$\begin{aligned} R_{SX}(\tau) &= R_{SS}(\tau) + R_{Sv}(\tau) \\ &= R_{SS}(\tau), \end{aligned}$$

$$\begin{aligned} R_{XX}(\tau) &= R_{SX}(\tau) + R_{vX}(\tau) \\ &= R_{SX}(\tau) + R_{vS}(\tau) + R_{vv}(\tau) \\ &= R_{SX}(\tau) + R_{vv}(\tau). \end{aligned}$$



Therefore,

$$S_{SX}(\omega) = S_{SS}(\omega),$$

$$S_{XX}(\omega) = S_{SS}(\omega) + S_{VV}(\omega).$$

By substituting the above equations,

$$H(\omega) = \frac{S_{SS}(\omega)}{S_{SS}(\omega) + S_{VV}(\omega)},$$

$$\text{MSE} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_{SS}(\omega)S_{VV}(\omega)}{S_{SS}(\omega) + S_{VV}(\omega)} d\omega.$$

Exercise

A zero-mean, wide-sense stationary (WSS) signal $S(t)$ with power spectral density (PSD) $S_{SS}(\omega) = \frac{2}{\omega^2 + 1}$ is corrupted by additive white noise $v(t)$ with a constant power spectral density $S_{vv}(\omega) = 1$.

The signal and noise are completely uncorrelated (orthogonal), so that $R_{Sv}(\tau) = 0$.

The noisy observed process is given by:

$$X(t) = S(t) + v(t)$$

- a) Find the transfer function $H(\omega)$ of the optimal non-causal Wiener filter to estimate $S(t)$ from $X(t)$.
- b) Calculate the resulting minimum MSE of this filter.

3. Prediction by Innovation Filter

■ Problem:

Using past state of $S(t - \tau)$, $\tau \in [0, \infty]$, estimate the future state of $S(t + \lambda)$ as

$$\hat{S}(t + \lambda) = \int_0^\infty h(\alpha) S(t - \alpha) d\alpha$$

Find the optimal filter $h(\alpha)$.

■ Solution:

By orthogonality principle,

$$S(t + \lambda) - \hat{S}(t + \lambda) \perp S(\xi) \text{ for all } \xi \in [-\infty, 0]$$

Denote $\xi = t - \tau$. For positive value of τ ($\tau \geq 0$),

$$E\left\{\left[S(t + \lambda) - \int_0^\infty h(\alpha)S(t - \alpha)d\alpha\right]S(t - \tau)\right\} = 0$$

$$E\{S(t + \lambda)S(t - \tau)\} \\ - \int_0^\infty h(\alpha)E\{S(t - \alpha)S(t - \tau)\}d\alpha = 0$$

$$R_{SS}(\tau + \lambda) - \int_0^\infty h(\alpha)R_{SS}(\tau - \alpha)d\alpha = 0$$

Solution $h(\alpha)$ of Wiener-Hopf integral equation gives
Causal Wiener Filter.

Because the integration range of τ is one-sided, transformation is not straightforward.

Solution by Innovation Filter

If stochastic process $S(t)$ is regular, it can be represented as output of innovation filter $L(s)$, to which white noise $I(t)$ is applied as input.

$$S(t + \lambda) = \int_0^\infty l(\alpha) I(t + \lambda - \alpha) d\alpha$$

Removing range $[0, \lambda]$, integration from past provides the optimal prediction.

$$\begin{aligned} \hat{S}(t + \lambda) &= \int_\lambda^\infty l(\alpha) I(t + \lambda - \alpha) d\alpha \\ &= \int_0^\infty l(\beta + \lambda) I(t - \beta) d\beta. \end{aligned}$$

■ Proof

Prediction error

$$S(t + \lambda) - \hat{S}(t + \lambda) = \int_0^\lambda l(\alpha) I(t + \lambda - \alpha) d\alpha$$

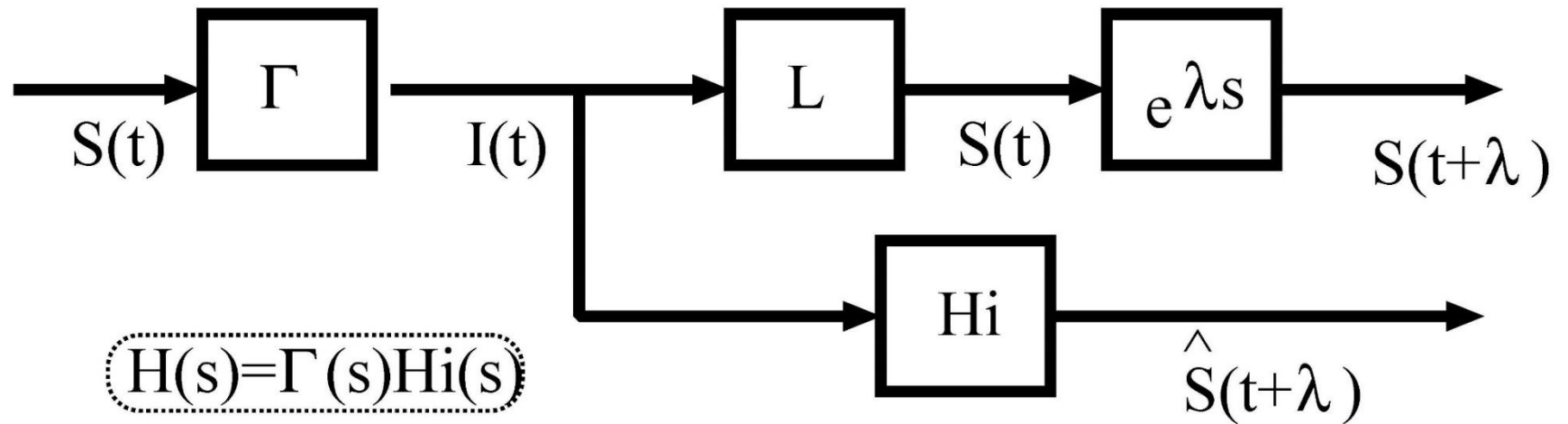
depends only upon interval $t \in (t, t + \lambda)$. They are orthogonal to past $I(t)$ or past $S(t)$.

■ Construction

Prediction $\hat{S}(t + \lambda)$ is a response of the following filter with white noise input $I(t)$:

$$h_i(t) = l(t + \lambda)U(t),$$

$$H_i(s) = \int_0^\infty h_i(t)e^{-st} dt.$$



Therefore, for input $S(t)$, corresponding filter is given by

$$H(s) = \frac{H_i(s)}{L(s)}$$

■ Procedure

1. Factorization: $S_{SS}(s) = L(s)L(-s)$
2. Find $l(t)$ by inverse Laplace transform of $L(s)$, and construct $h_i(t) = l(t + \lambda)U(t)$.
3. Find $H_i(s)$ by Laplace transform of $h_i(t)$, and determine $H(s) = H_i(s)/L(s)$.

■ Error

$$\begin{aligned}\text{MSE} &= E \left\{ \left| \int_0^\lambda l(\alpha) I(t + \lambda - \alpha) d\alpha \right|^2 \right\} \\ &= \int_0^\lambda l^2(\alpha) d\alpha\end{aligned}$$

Exercise

Suppose the autocorrelation function of a zero-mean WSS signal $S(t)$ is given by

$$R_{SS}(\tau) = 2\alpha e^{-\alpha|\tau|} \text{ where } \alpha > 0.$$

Construct its optimal causal prediction filter $H(s)$ to estimate $S(t + \lambda)$ for $\lambda > 0$.

Answer

1. Spectral Factorization

First, we find the Power Spectral Density $S_{SS}(s)$ by taking the bilateral Laplace transform of $R_{SS}(\tau)$:

$$S_{SS}(s) = \int_{-\infty}^{\infty} 2\alpha e^{-\alpha|\tau|} e^{-s\tau} d\tau = \frac{4\alpha^2}{\alpha^2 - s^2} = \left(\frac{2\alpha}{\alpha + s} \right) \left(\frac{2\alpha}{\alpha - s} \right)$$
$$L(s) = \frac{2\alpha}{\alpha + s}$$

2. Inverse Laplace Transform and Shifting

$$l(t) = \mathcal{L}^{-1} \left\{ \frac{2\alpha}{\alpha + s} \right\} = 2\alpha e^{-\alpha t} U(t)$$

We shift this response forward into the future by our look-ahead prediction window λ .

$$h_i(t) = l(t + \lambda)U(t) = 2\alpha e^{-\alpha(t+\lambda)} U(t) = 2\alpha e^{-\alpha\lambda} e^{-\alpha t} U(t)$$

3. Forward Laplace Transform

$$H_i(s) = \mathcal{L}\{2\alpha e^{-\alpha\lambda} e^{-\alpha t} U(t)\} = \frac{2\alpha e^{-\alpha\lambda}}{\alpha + s}$$

Hence,

$$H(s) = \frac{H_i(s)}{L(s)} = e^{-\alpha\lambda}.$$

By inverse Laplace transform, $h(t) = e^{-\alpha\lambda} \delta(t)$;
this leads to $\hat{S}(t + \lambda) = e^{-\alpha\lambda} S(t)$.

→ For prediction, only present state $S(t)$ is important;
past state has no influence.

4. Linear Prediction

■ **Problem** Given time series of $X[0], X[1], \dots, X[N-1]$, predict one-step future state $X[N]$ as

$$\hat{X}[N] = -\sum_{n=0}^{N-1} a_n X[n].$$

Find the optimal coefficients a .

■ **Solution** Error

$$\begin{aligned}\epsilon &= X[N] - \hat{X}[N] = X[N] + \sum_{n=0}^{N-1} a_n X[n] \\ &= a_0 X[0] + a_1 X[1] + \dots + a_{N-1} X[N-1] + X[N] \\ &= \sum_{n=0}^N a_n X[n] \quad (\text{where } a_N = 1)\end{aligned}$$

is orthogonal to data

$$\begin{aligned}E\{\epsilon X^*[k]\} &= \sum_{n=0}^N a_n E\{X[n] X^*[k]\} = 0 \\ &\quad (k = 0, 1, \dots, N-1)\end{aligned}$$

Supposing $X[n]$ is stationary,

$$E\{X[n]X^*[k]\} = R(n - k) = r_{n-k} = r_{k-n}^*$$

Therefore,

$$E\{\epsilon X^*[k]\} = \sum_{n=0}^N a_n r_{n-k} = 0,$$

$$a_N = 1, k = 0, 1, \dots, N - 1.$$

$$a_0 r_0 + a_1 r_1 + a_2 r_2 + \dots + a_{N-1} r_{N-1} + r_N = 0$$

$$a_0 r_1^* + a_1 r_0 + a_2 r_1 + \dots + a_{N-1} r_{N-2} + r_{N-1} = 0$$

$\vdots$

$$a_0 r_{N-1}^* + a_1 r_{N-2}^* + a_2 r_{N-3}^* + \dots + a_{N-1} r_0 + r_1 = 0$$



Minimum squared error is

$$\begin{aligned}\sigma^2 &= E\{|\epsilon|^2\} \\&= E\{\epsilon(X^*[N] + \sum_{n=0}^{N-1} a_n^* X^*[n])\} \\&= E\{\epsilon X^*[N]\} + \sum_{n=0}^{N-1} a_n^* E\{\epsilon X^*[n]\} \\&= E\{\epsilon X^*[N]\} \\&= E\{(\sum_{n=0}^N a_n X[n])X^*[N]\} \\&= \sum_{n=0}^N a_n E\{X[n]X^*[N]\} \\&= \sum_{n=0}^N a_n r_{N-n}^* \\&= a_0 r_N^* + a_1 r_{N-1}^* + a_2 r_{N-2}^* + \cdots + a_{N-1} r_1^* + r_0\end{aligned}$$

In matrix representation,

$$\begin{bmatrix} r_0 & r_1 & r_2 & \cdots & r_N \\ r_1^* & r_0 & r_1 & \cdots & r_{N-1} \\ r_2^* & r_1^* & r_0 & \cdots & r_{N-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{N-1}^* & r_{N-2}^* & \cdots & r_0 & r_1 \\ r_N^* & r_{N-1}^* & \cdots & r_1^* & r_0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{N-1} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ \sigma^2 \end{bmatrix}$$

$(N + 1) \times (N + 1)$ matrix on the left-hand side is called **Toeplitz** matrix.

Denoting the Toeplitz matrix as T and its inverse as

$$\mathbf{T}^{-1} = \begin{bmatrix} T_{0,0}^{-1} & T_{0,1}^{-1} & \cdots & T_{0,N}^{-1} \\ T_{1,0}^{-1} & T_{1,1}^{-1} & \cdots & T_{1,N}^{-1} \\ \vdots & \vdots & \ddots & \vdots \\ T_{N,0}^{-1} & T_{N,1}^{-1} & \cdots & T_{N,N}^{-1} \end{bmatrix}$$

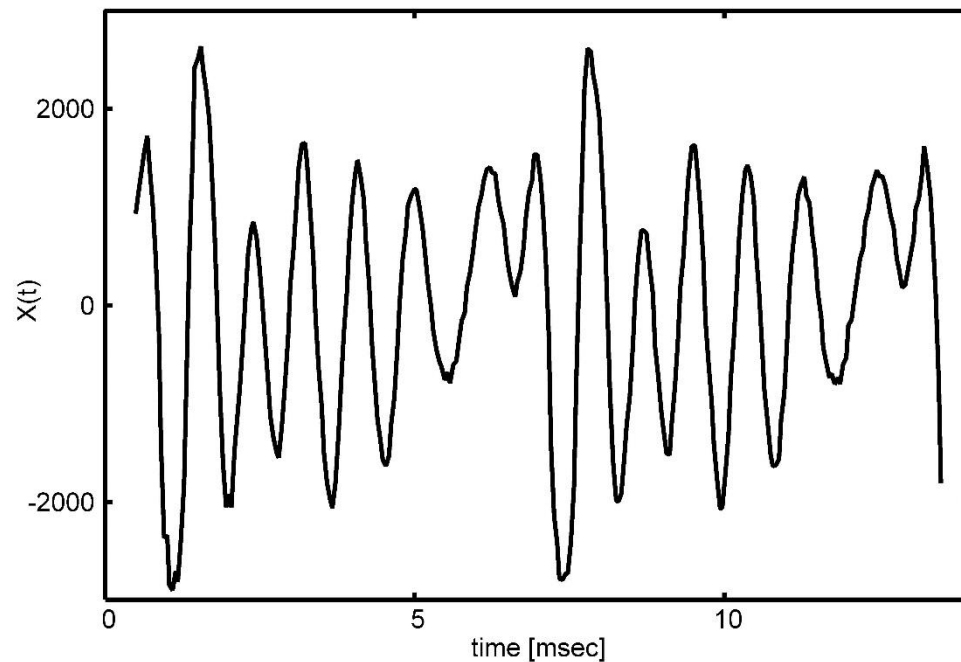
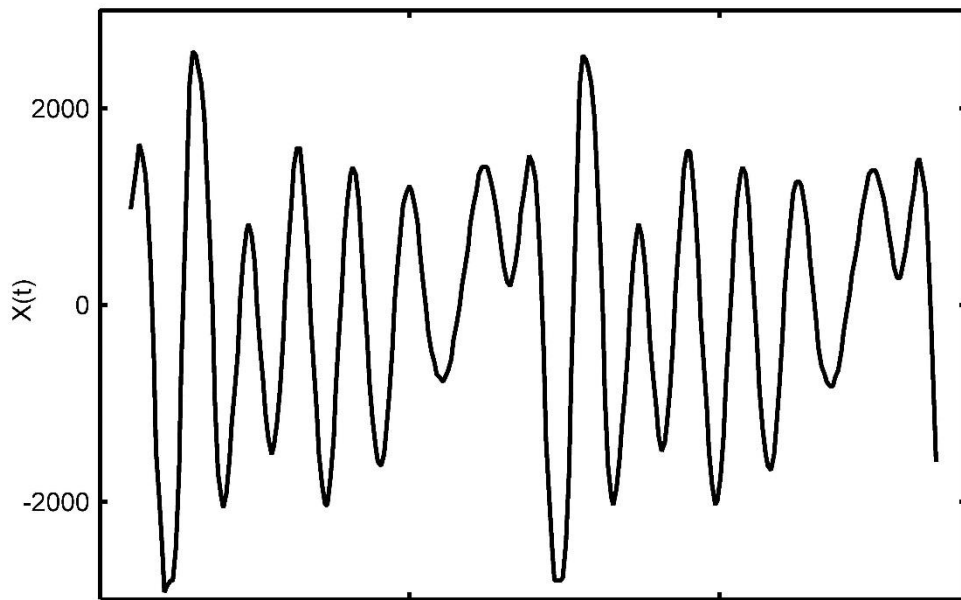
then

$$\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \\ 1 \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \sigma^2 \end{bmatrix} = \sigma^2 \begin{bmatrix} T_{0,N}^{-1} \\ T_{1,N}^{-1} \\ \vdots \\ T_{N-1,N}^{-1} \\ T_{N,N}^{-1} \end{bmatrix}$$

Therefore, coefficients a_i and minimum mean square error are given by

$$\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = \frac{1}{T_{N,N}^{-1}} \begin{bmatrix} T_{0,N}^{-1} \\ T_{1,N}^{-1} \\ \vdots \\ T_{N-1,N}^{-1} \end{bmatrix},$$

$$\sigma^2 = \frac{1}{T_{N,N}^{-1}}$$

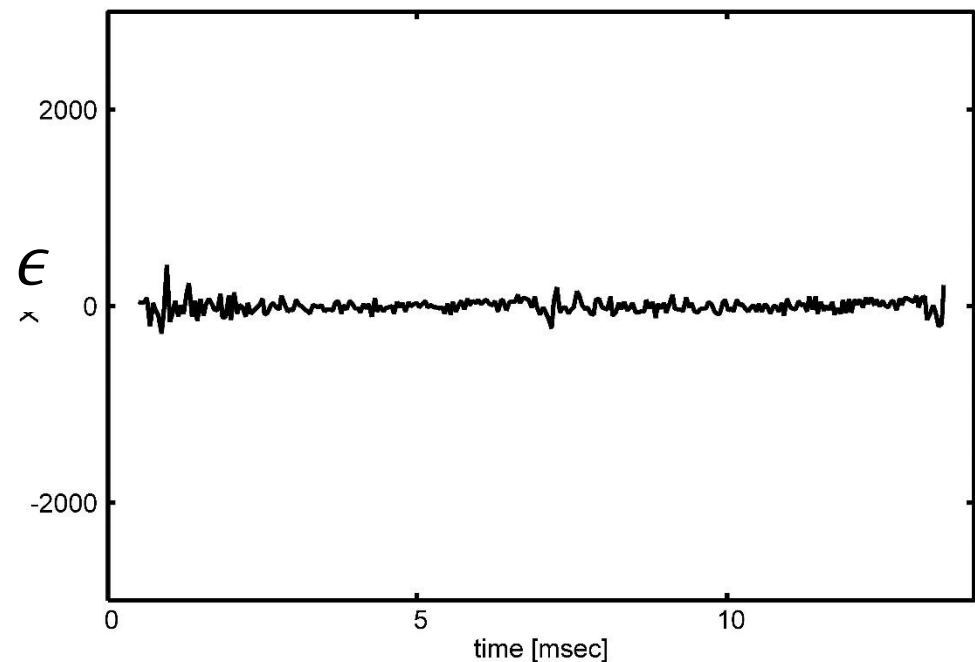


Linear prediction in speech signal

(left top: original)

(left bottom: LP)

(right: residue error)



5. Extension of LMMSE to Vector Form

Each element of random vector $\theta = \{\theta_1, \theta_2, \dots, \theta_p\}$ is estimated from N data $X[0], X[1], \dots, X[N-1]$ as

$$\theta_i = \sum_{n=0}^{N-1} a_{in} X[n] + a_{iN} \quad (i = 1, 2, \dots, p).$$

From the scalar version of LMMSE,

$$\begin{aligned} \hat{\theta}_i &= E(\theta_i) + C_{\theta_i X} C_{XX}^{-1} (X - E(X)), \\ Bmse(\hat{\theta}_i) &= C_{\theta_i \theta_i} - C_{\theta_i X} C_{XX}^{-1} C_{X \theta_i}. \end{aligned}$$

In vector representation,

$$\begin{aligned}\hat{\theta} &= \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} C_{\theta_1 X} \\ C_{\theta_2 X} \\ \vdots \\ C_{\theta_p X} \end{bmatrix} C_{XX}^{-1} (\mathbf{X} - E(\mathbf{X})) \\ &= E(\theta) + \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} (\mathbf{X} - E(\mathbf{X})).\end{aligned}$$

$$\begin{aligned}\text{Error matrix: } M_{\hat{\theta}} &= E \left((\theta - \hat{\theta})(\theta - \hat{\theta})^T \right) \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta X} \mathbf{C}_{XX}^{-1} \mathbf{C}_{X\theta},\end{aligned}$$

$$\text{MMSE: } Bmse(\hat{\theta}_i) = [M_{\hat{\theta}}]_{ii}$$

6. Sequential LMMSE Estimation

From $N - 1$ random data $X[1], X[2], \dots, X[N - 1]$, we estimate random vector θ via LMMSE estimation, where the estimate and the error matrix are given by

$$\hat{\theta}[N - 1],$$

$$M[N - 1] = E \left[(\theta - \hat{\theta}[N - 1])(\theta - \hat{\theta}[N - 1])^T \right].$$

Suppose now that N th data $X[N]$ is newly observed, where observation matrix is decomposed as

$$H[N] = \begin{bmatrix} H[N - 1] \\ h^T[N] \end{bmatrix} = \begin{bmatrix} (N - 1) \times p \\ 1 \times p \end{bmatrix}.$$



Then, the estimate and the error matrix can be updated as follows.

$$\begin{aligned}\hat{\theta}[n] &= \hat{\theta}[n-1]K[n](x[n] - h^T[n]\hat{\theta}[n-1]), \\ K[n] &= \frac{M[n-1]h[n]}{\sigma_n^2 + h^T[n]M[n-1]h[n]}, \\ M[n] &= (I - K[n]h^T[n])M[n-1].\end{aligned}$$

Here, we supposed that the errors $\theta - \hat{\theta}[n]$ are independent and their covariance matrix has diagonal elements of $\text{diag}(\sigma_0^2, \sigma_1^2, \dots, \sigma_n^2)$.

Exercise

We want to estimate a parameter vector $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$. Our prior knowledge tells us that its mean is zero, $E(\boldsymbol{\theta}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, and its prior error covariance matrix is: $\mathbf{M}[0] = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$.

We receive a first scalar observation $x[1]$ via the linear model:

$$x[1] = \mathbf{h}^T[1]\boldsymbol{\theta} + w[1]$$

The observation row vector is $\mathbf{h}^T[1] = [1 \quad 1]$.

The measurement noise $w[1]$ is zero-mean with variance $\sigma_1^2 = 1$, and is independent of $\boldsymbol{\theta}$.

1. Calculate the Kalman Gain vector $\mathbf{K}[1]$.
2. Assuming the actual observed value is $x[1] = 3$, find the updated vector estimate $\hat{\boldsymbol{\theta}}[1]$.
3. Calculate the updated error covariance matrix $\mathbf{M}[1]$.

Answer

1. Kalman Gain vector $K[1]$:

$$K[1] = \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}$$

2. The updated vector estimate $\hat{\theta}[1]$:

$$\hat{\theta}[1] = \begin{bmatrix} 1.2 \\ 1.2 \end{bmatrix}$$

3. The updated error covariance matrix $M[1]$:

$$M[1] = \begin{bmatrix} 1.2 & -0.8 \\ -0.8 & 1.2 \end{bmatrix}$$

Batch LMMSE vs Sequential LMMSE

Metric	Batch LMMSE	Sequential LMMSE
Data Needed	Must keep and process all N data points simultaneously.	Needs only the current data point $x[n]$ and the previous state $(\hat{\theta}[n-1], M[n-1])$.
Matrix Inversion	Requires inverting an $N \times N$ data covariance matrix (C_{XX}^{-1}), which scales horribly ($O(N^3)$).	The denominator in the gain calculation is a scalar ($\sigma_n^2 + h^T M h$). Inversion becomes a simple division!
Memory	High (grows with time).	Low and constant (ideal for real-time embedded systems).